Assignment 8

22/05/26

Question 1

Matrix Construction & Graph Metrics

Adjacency matrix (A) & Degree Matrix (D)

Let the nodes be ordered as A, B, C, D, E, F

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 \end{bmatrix}$$

Node degree (K_i)

- A = 2
- B = 3
- C = 3
- D = 2
- E = 3
- F = 3

Graph Density & Degree Av.

Total Nodes - (N) = 6

Total Edges - (E) = 8

Graph Density = 0.533

Av. Degree = 2.67

Clustering Coefficient (L_i)

$$L_i = \frac{2e_i}{K_i(K_i - 1)}$$

A = 0/1

B = 0/3

C = 1/3

C = 0/1

D = 0/1

E = 0/1

F = 0/3